Reg. No. | | | | | | | | | | | [MSCCH202]

SECOND SEMESTER UNIVERSITY EXAMINATION, JUNE 2016

M.Sc. (CHEMISTRY)

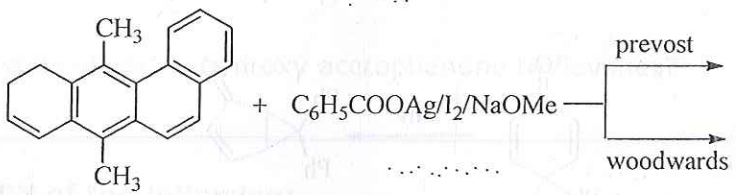
ORGANIC CHEMISTRY-II

Time : 03 Hours **Maximum Marks:80**

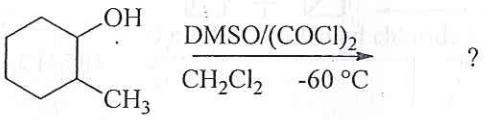
Instructions: **1.** Questions 1 to 6 carry 15 marks each, answer any four.

2. Question 7 is compulsory, carries 20 marks.

1. **a.** Predict the product and explain the mechanism for the following reaction.



b. Predict the product and explain the mechanism for the following reaction



10+5

2. **a.** Write a note on Adam's catalyst.

b. Discuss the mechanism for the hydrogenation of alkenes by heterogeneous catalysts.

c. Explain the steroselectivity of hydride reduction on the basis of cyclohexanone derivatives. **5+5+5**

3 **a.** Write a detailed description of intramolecular and intermolecular hydrogen abstractions involved in n-π* transitions with two appropriate examples for each. Explain their mechanisms.

b. What is a Norrish type-I reaction? Explain its reaction mechanism with a non symmetrical acyclic ketone as an example. **10+5**

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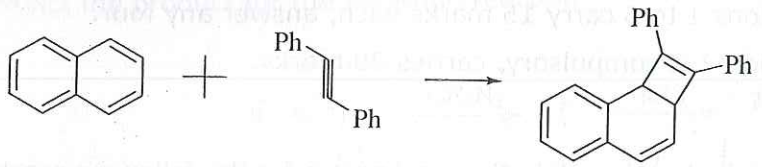
4. a. Suggest a mechanism for any three of the following dimerization/ cycloaddition/ rearrangement/sensitization reactions involved in π-π* transitions.

i.



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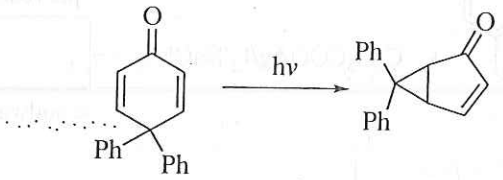

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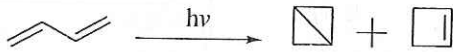

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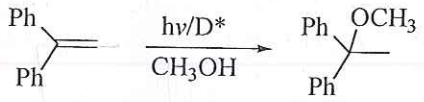
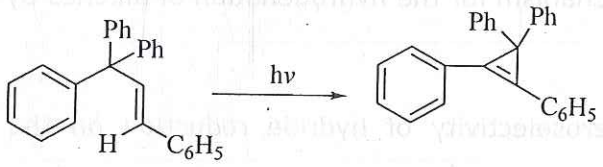


v.



b. Write a note on photo Fries rearrangement. **10+5**

5. a. Explain any five methods for the synthesis of epoxides.

b. Discuss the ring-opening reactions of oxetane.

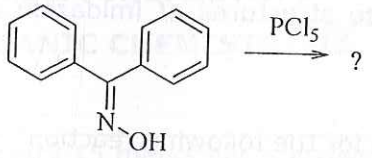
c. Outline the steps involved in the synthesis of the following heterocycles by the methods indicated below,

i. Oxazole (Robinson-Gabriel method)

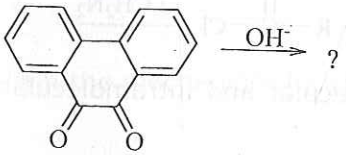
ii. Pyrazole (3+2 approach) **5+4+6**

6. a. Predict the product for the following reactions and explain the mechanism

i.



ii.



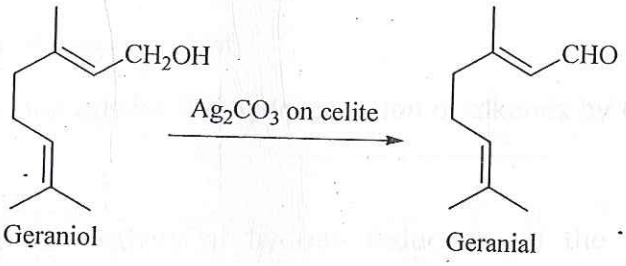
b. How do you convert 2-hydroxy acetophenone to flavones? **10+5**

7. **Answer any TEN of the following:** **10×2=20**

a. Predict the product for the following reaction with mechanism

$$CH_3(CH_2)_6CH_2OH \xrightarrow[\text{ii) DMSO}]{\text{i) } p\text{-toluene sulfonyl chloride}} \quad ?$$

b. Write the mechanism for the following conversion



c. What is homogenous hydrogenation? What are its special uses compared to heterogenous hydrogenation?

d. Write a note on Meerwein-Pondorff-Verley reduction.

e. Illustrate a singlet state and a triplet state with an energy level diagram.

f. Calculate the energy of one mole of photons having a wavelength of 700 nm.

g. Illustrate an *ene* reaction of singlet oxygen.

h. Predict the possible photochemical product for,

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i. Describe any two methods each for the synthesis of thiirane.

j. Write the resonance structures of imidazole and discuss the point of electrophilic substitution.

k. Predict the product for the following reaction

$$R-\overset{\overset{\displaystyle O}{\|}}{C}-Cl \xrightarrow{2\,CH_2N_2} \; ? \xrightarrow{Ag_2O} \; ?$$

l. What are intermolecular and intramolecular rearrangements? Give one example for each.